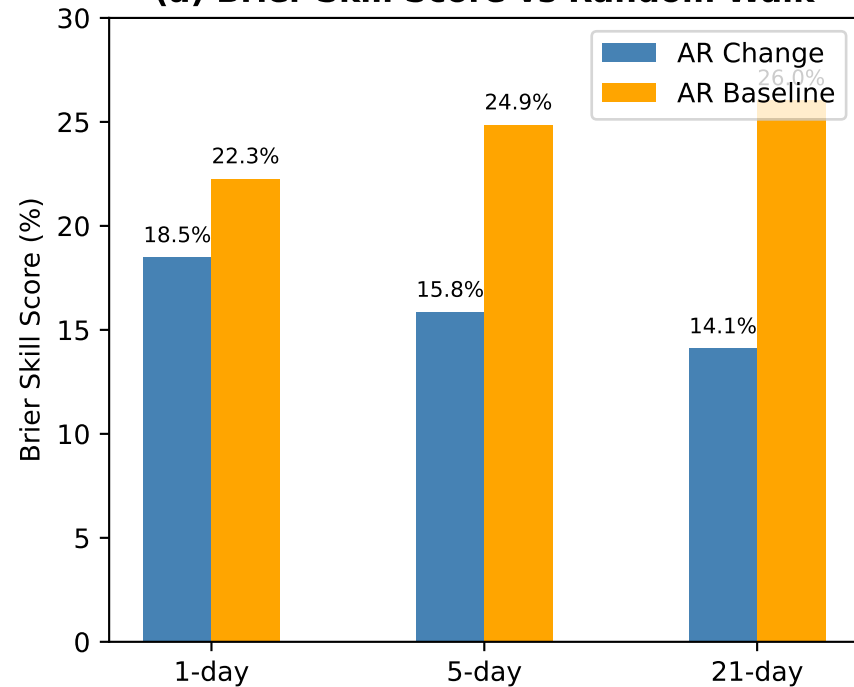
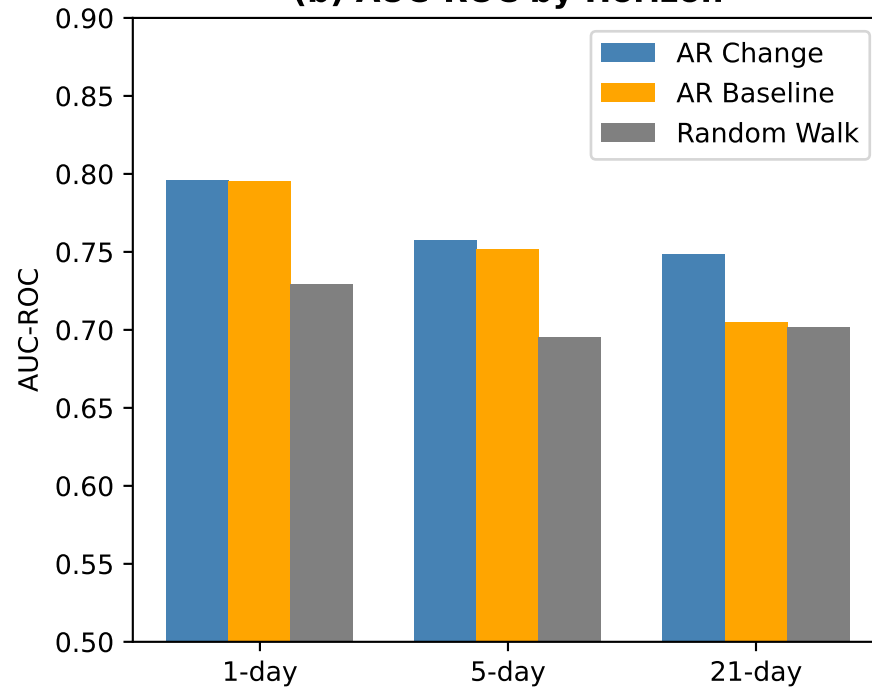
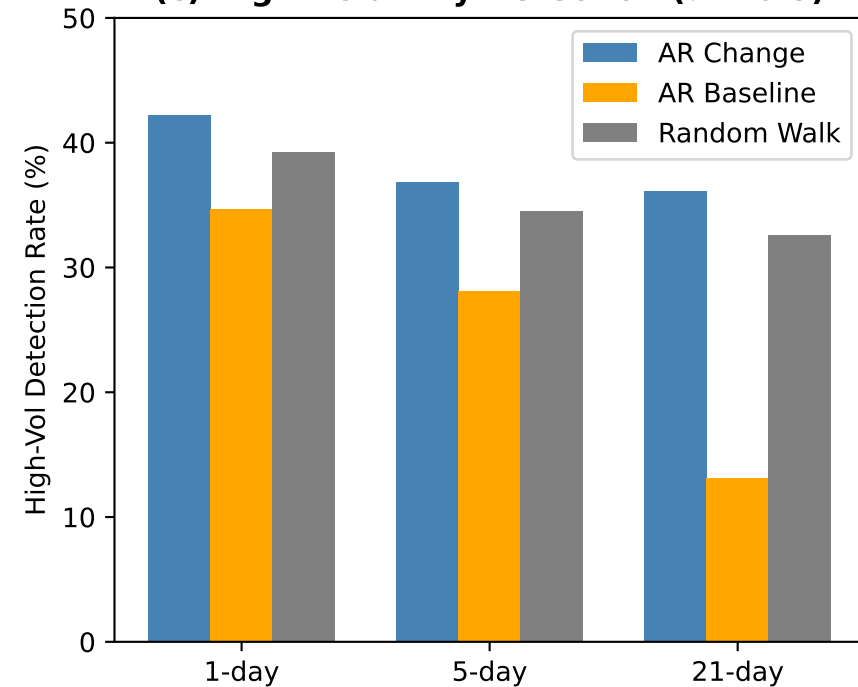


(a) Brier Skill Score vs Random Walk**(b) AUC-ROC by Horizon****(c) High-Volatility Detection ($\theta = 0.5$)**

Note: computed on the one-sided $K=2$ regime-probability series (no look-ahead); $n_{\text{forecasts}} = 4748$ per horizon. Always-calm hit-rate baseline: 1d 85.7% (AR Change -0.9pp), 5d 85.7% (AR Change -2.4pp), 21d 85.7% (AR Change -2.9pp).